L29 Introduction to the Phase Plane

$my'' + by' + ky = 0$

Section 5.4

$\Rightarrow y'' = -\frac{b}{m}y' - \frac{k}{m}y$

Consider a system of two first-order DEs in the form

set $v = y'$, $v' = y''$

$$\begin{cases} \frac{dx}{dt} &= f(x,y) \\ \frac{dy}{dt} &= g(x,y) \end{cases} \quad \leftarrow \text{no } t\text{'s}$$

$$\begin{cases} y' = v \\ v' = -\frac{b}{m}v - \frac{k}{m}y \end{cases}$$

Such a system is called **autonomous** because the independent variable t does not appear on the right-hand side.

If one $x(t), y(t)$ pair of functions solves the system, so will the time-shifted pair $x(t+c), y(t+c)$ for any constant c.

A solution to the autonomous system is a pair of functions $x(t)$, $y(t)$ that satisfies the system for all t in some interval I.

We can visualize the system as a pair of graphs.We can also plot this information by charting the path traced out by the coordinate pair $(x(t),\, y(t))$ as t increases.

Def. If $(x(t), y(t))$ is a solution pair to * for t in the interval I, then a plot in the $xy-$ plane of the parameterized curve$x = x(t)$ $y = y(t)$ with arrows indicating the direction of increasing t is said to be the **trajectory** of the system. We call the $xy-$ plane the **phase plane** of the system which can be represented by a set of trajectories in this plane as the **phase portrait**.

$$\begin{cases} \frac{dx}{dt} = f \\ \frac{dy}{dt} = g \end{cases}$$

Since t does not appear on the RHS, we can divide the two equations to get

M-S example

$$\frac{dy}{dx} = \frac{\cancel{\frac{dy}{dt}}}{\cancel{\frac{dx}{dt}}} = \frac{g(x,y)}{f(x,y)}$$

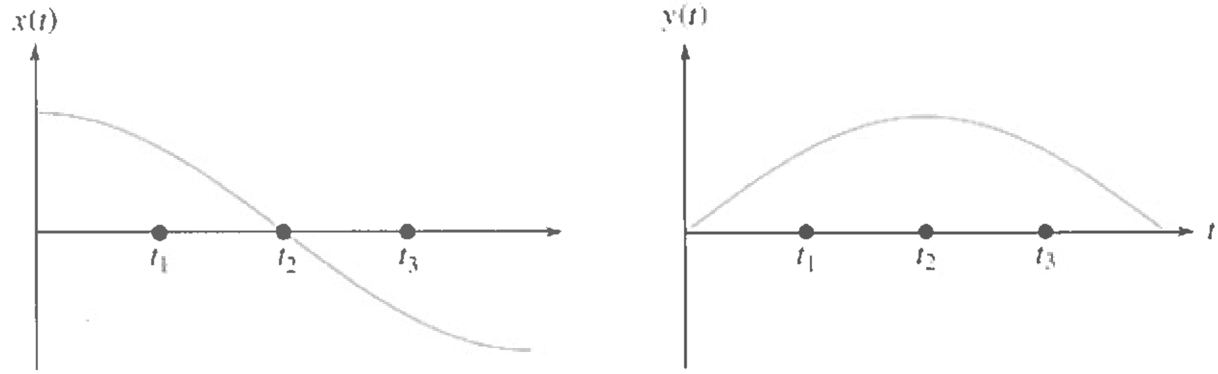
$$\frac{dv}{dy} = \frac{-\frac{b}{m}v - \frac{k}{m}y}{v} = -\frac{b}{m} - \frac{k}{m}\frac{y}{v}$$

homogeneous
1st order

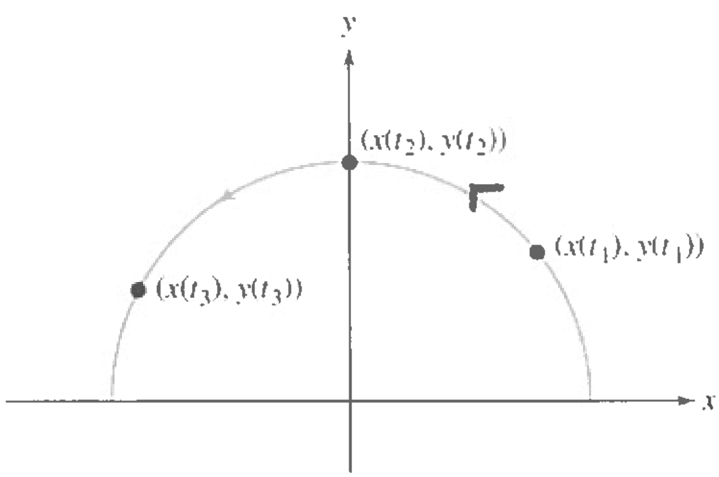
We refer to this as the **phase-plane equation.**

We can use the method of direction fields (section 1.3) to visualize the solution graphs.

A solution to our original problem is a pair of functions of t. We can visualize them as a pair of graphs



or we could plot the points $(x(t), y(t))$ to get the trajectory of the solution pair in the xy-plane



$$\begin{cases} \frac{dx}{dt} = f(x,y) \\ \frac{dy}{dt} = g(x,y) \end{cases}$$

Def. A point (x_0, y_0) where $f(x_0, y_0) = 0$ and $g(x_0, y_0) = 0$ is called a **critical point** or an **equilibrium point** of the system. The corresponding constant solution $x(t) = x_0,\ y(t) = y_0$ is called an **equilibrium solution**. The set of all critical points is called the **critical point set**.

ex. Sketch the direction field in the phase plane for the system below and identify its critical point.

$$\begin{cases} \frac{dx}{dt} &= -x \\ \frac{dy}{dt} &= -2y \end{cases} \Rightarrow \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-2y}{-x} \quad \text{(phase plane equation)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2y}{x} \Rightarrow \int \frac{1}{y}\,dy = \int \frac{2}{x}\,dx \Rightarrow \ln|y| = 2\ln|x| + C$$

separable

$$\Rightarrow \ln|y| = \ln|x^2| + C$$

$$\Rightarrow y = Cx^2$$

critical point: $\begin{cases} -x = 0 \\ -2y = 0 \end{cases} \Rightarrow \begin{matrix} x = 0 \\ y = 0 \end{matrix}$

$x = 1$, $y = 1$

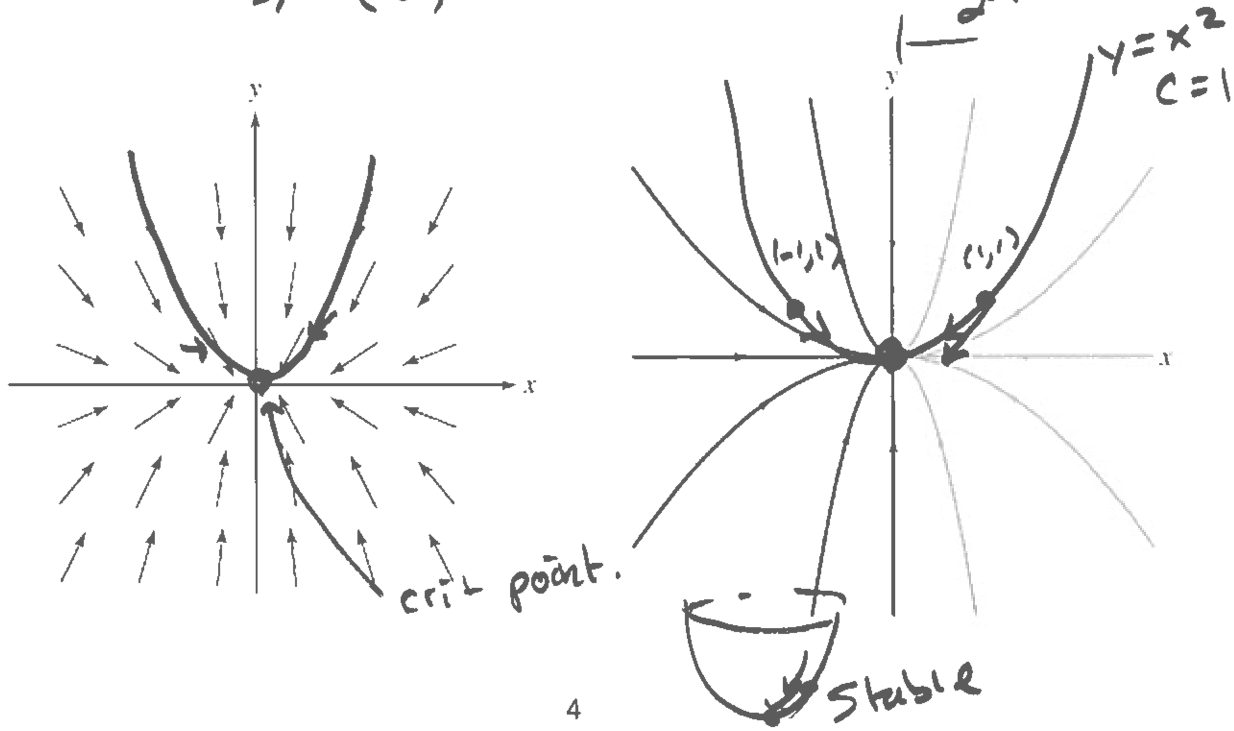
$x' = -1$

$y' = -2$

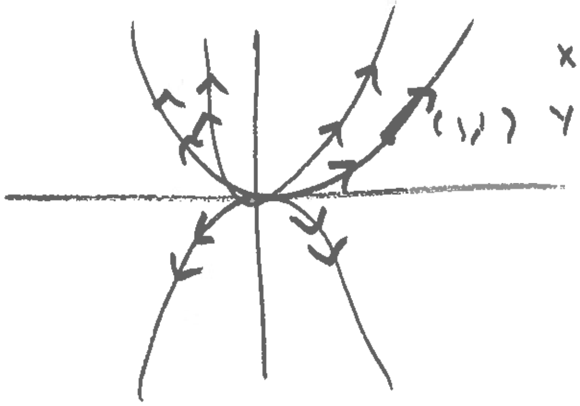
$$\begin{cases} \frac{dx}{dt} = -x \\ \frac{dy}{dt} = -2y \end{cases}$$

trajectories are $y = Cx^2$

$\Rightarrow (t, Ct^2)$



Ex:
$$\begin{cases} dx/dt = x \\ dy/dt = 2y \end{cases} \Rightarrow \frac{dy}{dx} = \frac{2y}{x} \Rightarrow y = Cx^2$$



ex. Sketch the direction field in the phase plane for the system below and identify its critical point.

$$\begin{cases} \frac{dx}{dt} &= 5x-3y-2 \\ \frac{dy}{dt} &= 4x-3y-1 \end{cases}$$

critical point(s):

$$\begin{cases} 5x-3y-2=0 & (1) \\ 4x-3y-1=0 & (2) \end{cases}$$

$(1)-(2): (5x-2)-(4x-1)=0$

$\Rightarrow x-1=0 \Rightarrow \underline{x=1}$

in (1) $x=1$: $5(1)-3y-2=0$

$\Rightarrow -3y+3=0$

$\underline{y=1}$

$(1,1)$ critical point.

$$\frac{dy}{dx} = \frac{4x-3y-1}{5x-3y-2}$$

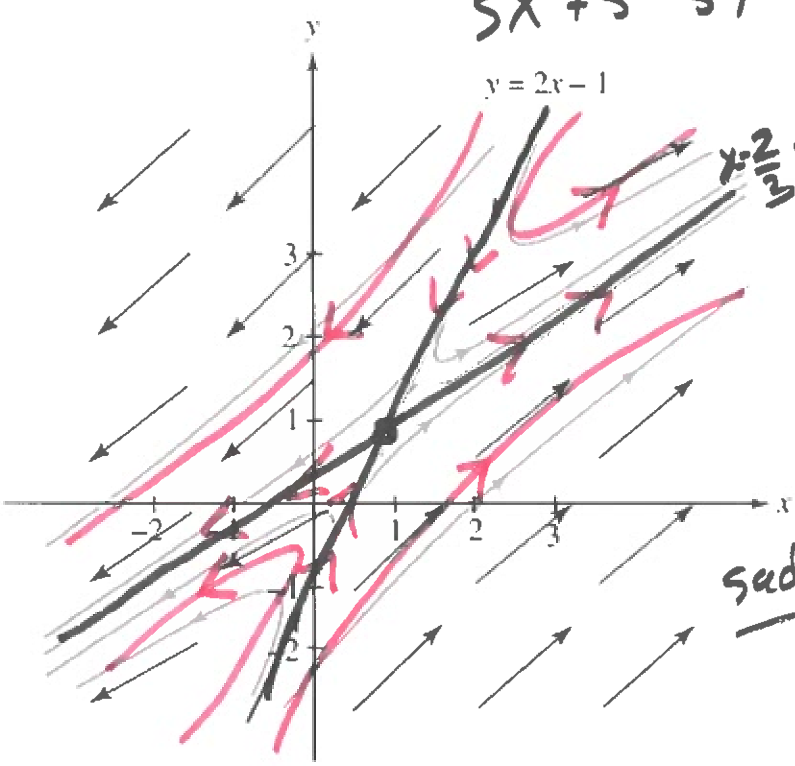
set $X+1=x$, $Y+1=y$

$$\frac{dY}{dX} = \frac{4(X+1)-3(Y+1)-1}{5(X+1)-3(Y+1)-2}$$

$$= \frac{4X+4-3Y-3-1}{5X+5-3Y-3-2} = \frac{4X-3Y}{5X-3Y}$$

homogeneous

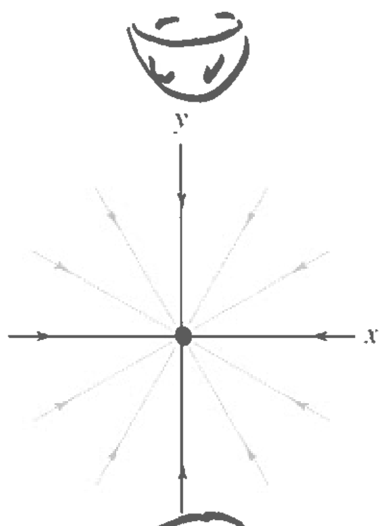
$$\ln\left|\frac{\left(3\frac{y-1}{x-1}-2\right)^{3/4}}{\left(\frac{y-1}{x-1}-2\right)^{1/4}}\right| = \ln(x-1) + C$$



saddle

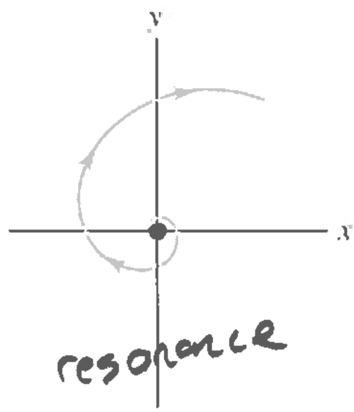
Theorem: End points are critical points. Let the pair $x(t), y(t)$ be a solution on $[0, \infty)$ to an autonomous system $\frac{dx}{dt} = f(x,y)$, $\frac{dy}{dt} = g(x,y)$ where f and g are continuous on the plane. If the limits $x^* = \lim_{t\to\infty} x(t)$ and $y^* = \lim_{t\to\infty} y(t)$ exist and are finite, then the point $(x^*, y*)$ is a critical point of the system.

Examples of different trajectory behaviors near critical points at the origin are given below.

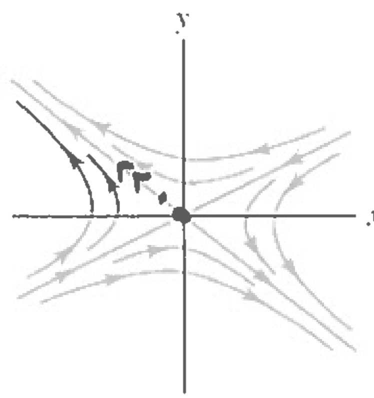


Node
(asymptotically stable)

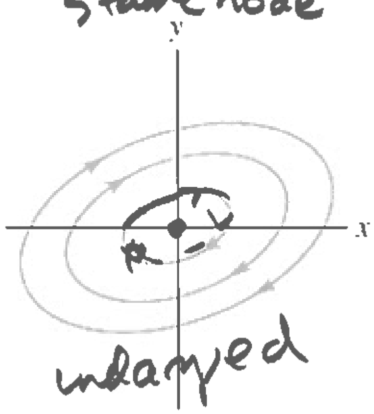
stable node



Spiral
(unstable)

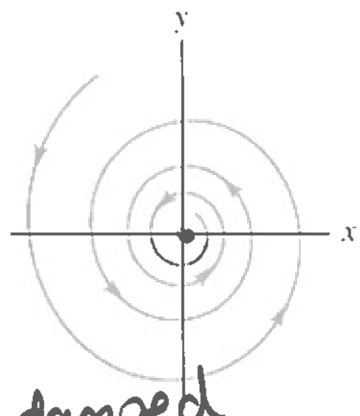


Saddle
(unstable)

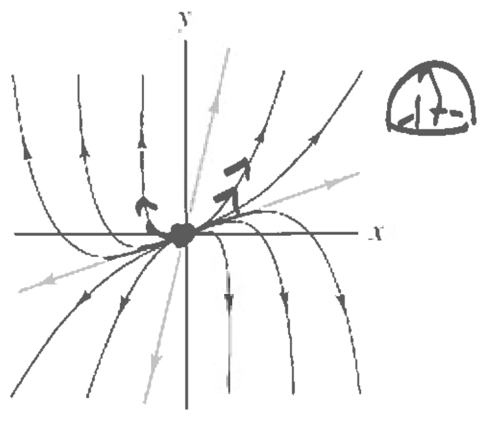


Center
(stable)

damped



Spiral
(asymptotically stable)



Node
(unstable)

Unstable critical points have **trajectories moving away** from them.

Stable critical points either have

- **trajectories moving towards** them (**asymptotically stable** critical points)
- **or circulating around** them (**stable** critical points).

Consider a second-order DE $y'' = f(y, y')$. We can convert it to a system of first-order equations by introducing $v = \frac{dy}{dt}$. Then we have

$$\begin{cases} \frac{dy}{dt} & = v \\ \frac{dv}{dt} & = f(y, v) \end{cases}$$

This allows us to analyze the behavior of an autonomous system by studying its phase plane diagram in the yv-plane.

ex. Sketch the direction field in the phase plane for

$$my'' + ky = 0. \quad \to \quad v = y' \Rightarrow v' = -\frac{k}{m}y$$

The system takes the form

$$\begin{cases} \frac{dy}{dt} &= v \\ \frac{dv}{dt} &= -\frac{ky}{m} \end{cases} \Rightarrow \frac{dv}{dy} = \frac{-\frac{k}{m}y}{v} \quad \text{separable}$$

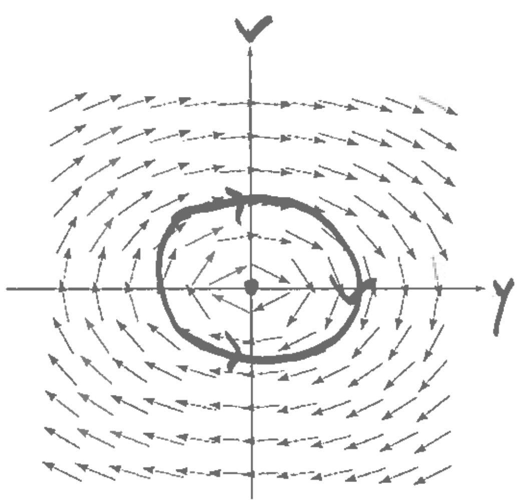
$v = 0$

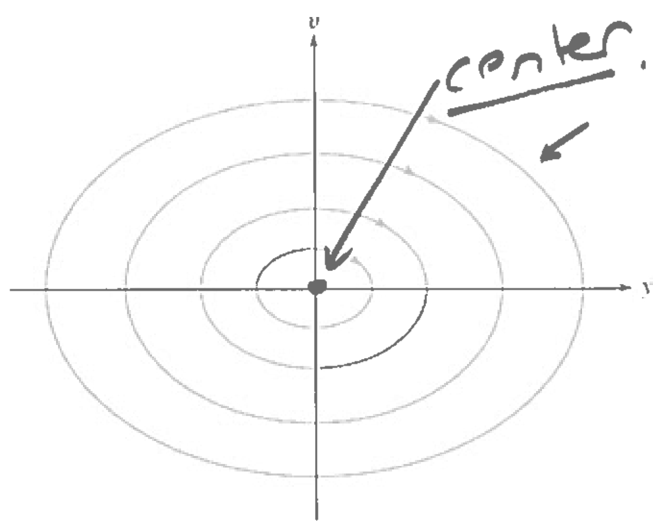
$-\frac{k}{m}y = 0 \Rightarrow (0,0)$

$$\Rightarrow \int mv\,dv = \int -ky\,dy$$

$$\Rightarrow \left(\frac{1}{2}mv^2 = -\frac{1}{2}ky^2 + C\right)\cdot 2$$

$$\Rightarrow mv^2 + ky^2 = C \quad \text{ellipses}$$

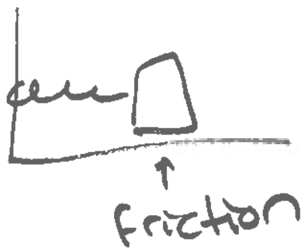




Application: dampered mass-spring oscillator

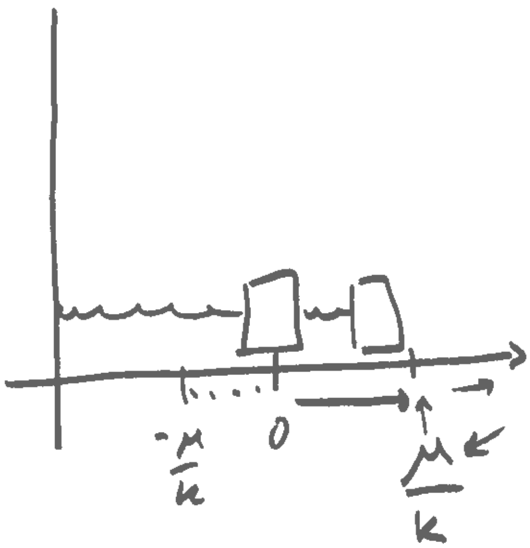
$$my'' + by' + ky = 0$$

$$\Rightarrow my'' = -ky \underbrace{-by'}_{\text{damp}} = -ky + F_{damp}$$

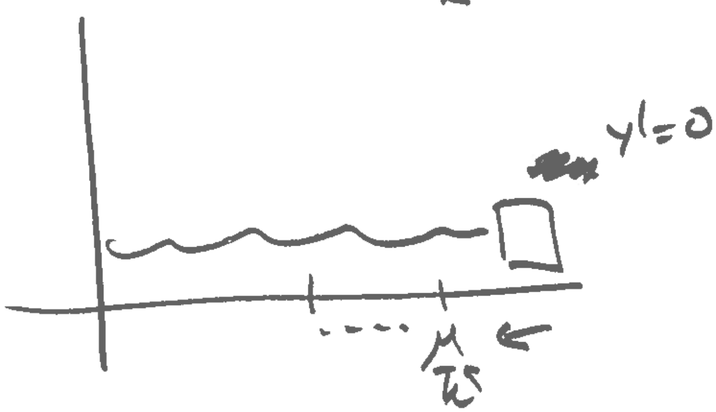


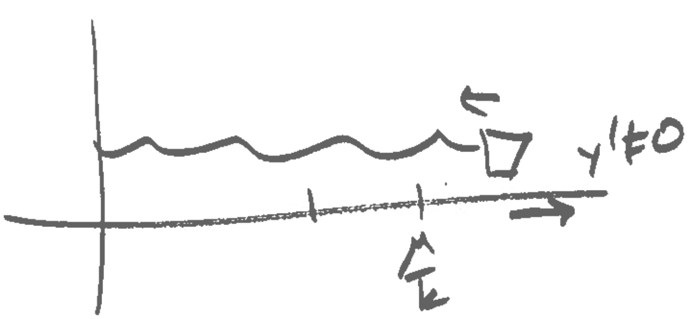
Instead, we can use

$$F_{friction} = \begin{cases} ky & |y| < \frac{\mu}{k},\ y' = 0 \\ -\mu\,\text{sign}(y) & ,\ |y| > \frac{\mu}{k},\ y' = 0 \\ \mu\,\text{sign}(y') & ,\ y' \neq 0 \end{cases}$$



$$\begin{cases} \text{sign}(\pi) = 1 \\ \text{sign}(-1) = -1 \\ \text{sign}(-3) = -1 \end{cases}$$





In the case where $my'' = -ky + F_{friction}$ $\quad m = \mu = k = 1$

$$\Rightarrow \quad y'' = -y + F_{friction} \qquad F_{friction} = \begin{cases} y, & |y| < 1,\ y' = 0 \\ \text{sign}(y) & y' = 0,\ |y| > 1 \\ -\text{sign}(v), & y' \neq 0 \end{cases}$$

If $y' \neq 0$ movement

$$y'' = -y + (-\text{sign}(v))$$

$$\Rightarrow \begin{cases} y' = v \\ v' = -y - \text{sign}(v) \end{cases}$$

$$\Rightarrow \frac{dv}{dy} = \frac{-y - \text{sign}(v)}{v} = \frac{-y \pm 1}{v} \quad \text{separable}$$

$$\Rightarrow \int v\,dv = \int (-y \pm 1)\,dy \Rightarrow \frac{1}{2}v^2 = -\frac{1}{2}(y \pm 1)^2 + C$$

$$\Rightarrow v^2 + (y \pm 1)^2 = C \quad \underline{\text{two circles}}.$$